

Article

Conditional Effectiveness of Volatility-Adaptive Exit Rules in Algorithmic Trading Systems: Evidence from the USD/JPY Market

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Abstract

This study examines Average True Range (ATR)-based take-profit (TP) and stop-loss (SL) rules as components of adaptive risk-management design within robust algorithmic trading systems in the USD/JPY foreign exchange market. Rather than simply evaluating whether volatility-adaptive exit rules improve trading performance, the study investigates the conditions under which they contribute to trading outcomes. To this end, the study adopts a systematic framework combining a broad Moving Average Convergence Divergence (MACD) parameter space with ATR-based TP/SL multiplier settings and distinguishes heterogeneous outcome patterns before and after ATR implementation. The results show that the effectiveness of ATR-based exit rules is conditional. Performance improvements occur only for specific combinations of model structures, exit-rule specifications, and market conditions. A notable finding is that optimizing trading-model parameters plays a primary role in determining the effectiveness of exit rules, while ATR-based exit rules function as complementary components that reinforce well-specified trading models. Moreover, under mildly mean-reverting market conditions, profit-enhancement cases expand across a broader range of optimized MACD parameter configurations, suggesting a previously underexplored interaction between market dynamics, model structure, and volatility-adaptive exit-rule design. Overall, the findings provide new evidence that the effectiveness of adaptive risk-management mechanisms depends on the interaction between model structure, exit-rule design, and market conditions, thereby offering broader insights into adaptive trading-system design under changing market conditions. For practitioners, the results suggest that adaptive ATR-based exit rules are most effective when combined with appropriately optimized trading models.

Keywords: algorithmic trading systems; adaptive risk management; volatility-adaptive exit rules; ATR; MACD; parameter optimization; foreign exchange market; USD/JPY



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1. Introduction

In financial markets, the possibility of predicting asset prices to some extent has long motivated the development of systematic and rule-based approaches to investment decision-making under uncertainty (Lo & MacKinlay, 1999). As electronic trading technologies became increasingly accessible, systematic and rule-based approaches were more readily implemented through automated, or algorithmic, trading systems. Their widespread adoption has contributed to improvements in market liquidity and reductions in transaction costs (Jain, 2005). Consequently, trading execution has increasingly shifted from manual procedures to automated systems (Hendershott & Moulton, 2007). Compared

with discretionary trading, algorithmic systems offer several structural advantages, including faster execution, enhanced information-processing capacity, and reduced exposure to behavioral biases such as fear and overreaction. More recently, algorithmic trading has continued to evolve with advances in computational technologies and data-driven trading methodologies, and it remains an active area of both academic research and practical application (Carè & Cumming, 2024).

Within this context, a growing body of literature has examined the impact of algorithmic trading on market liquidity and trading performance. For example, Hendershott et al. (2011) show that algorithmic trading improves market liquidity and enhances the incorporation of information into prices, thereby improving price efficiency, while Menkveld (2013) finds that high-frequency trading contributes to market liquidity provision. Cheung and Chinn (2001) and Luk and Smith (2001) provide early evidence that technical trading rules can generate excess returns in foreign exchange markets.

However, as algorithmic trading systems increasingly shift decision-making away from human discretion, trading outcomes become more dependent on system design rather than on continuous human intervention. In this context, adaptive risk-management mechanisms designed to respond to market price fluctuations—particularly those governing downside risk and trade exits—have become increasingly important components of trading systems, especially in foreign exchange markets characterized by high exchange-rate volatility.

Among these mechanisms, considerable attention has been devoted to the design and effectiveness of take-profit (TP) and stop-loss (SL) rules. These rules define exit conditions by specifying profit targets and loss limits, thereby shaping the trade-off between risk control and return realization. Existing studies on exit mechanisms can be broadly classified into two categories: (1) fixed exit rules, defined at constant price levels or as proportions of the entry price, and (2) volatility-adaptive exit rules, in which exit thresholds adjust according to market volatility.

Several studies illustrate the use of fixed exit-rule approaches. Consistent with Glynn and Iglehart (1995), who model exit rules as risk-control mechanisms, Krishnan and Menon (2009) employ a constant take-profit level of 20 pips in a trading system incorporating multiple technical indicators. Other studies examine multiple fixed exit thresholds. For example, Azzini and Tettamanzi (2008) employ staggered take-profit levels ranging from 0.33% to 0.67%, while Martinez et al. (2009) investigate stop-loss thresholds ranging from 0.1% to 2.0%. In addition, Bolgün et al. (2010) set take-profit and stop-loss levels at 3% and 2% of the entry price, respectively. More recently, Kang (2026) examines multiple fixed TP and SL rules defined as percentages of the entry price and reports that specific TP/SL combinations can improve risk-adjusted performance under optimized MACD parameter settings in the gold market.

Examples of volatility-adaptive exit-rule approaches include Chevallier et al. (2012), who employ a dynamic stop-loss rule based on returns relative to market variability over a specified number of fluctuation days, and Klement (2013), who defines stop-loss levels in units of the annual standard deviation of returns. Kaminski and Lo (2014) examine dynamic stop-loss and re-entry rules based on deviations from mean returns across multiple time horizons. Cervelló-Royo et al. (2015) define exit thresholds relative to the price range between the highest and lowest prices over a given period. Among these approaches, Vezeris et al. (2018) investigate ATR-based take-profit and stop-loss strategies across multiple asset classes, including foreign exchange, metals, energy, and cryptocurrencies.

Several studies report that stop-loss and take-profit rules can improve trading performance. However, despite these contributions, the existing literature exhibits several important limitations. First, many studies rely on limited combinations of trading-model

parameters and TP/SL settings, making it difficult to evaluate how the effectiveness of exit rules varies across a broader parameter space and different TP/SL settings. Second, prior studies focus primarily on aggregate or average-performance comparisons, obscuring heterogeneous outcomes across individual model configurations and making it difficult to distinguish performance improvements from deteriorations at that level. Third, relatively little attention has been paid to the interaction between model structure, exit-rule design, and market conditions in determining whether exit rules generate performance improvements. As a result, it remains unclear under what conditions exit-rule-based strategies improve performance and how their effectiveness depends on model structure, exit-rule design, and market conditions.

Such uncertainty limits the ability to design rule-based algorithmic trading systems capable of effectively balancing risk control and return realization. Trading performance depends not only on entry decisions but also on how positions are managed after they are opened. More broadly, overall trading system performance is determined by the interaction between model structure, exit-rule design, and prevailing market conditions.

Motivated by these limitations and the growing importance of robust algorithmic trading-system design, this study aims to derive design-relevant insights for rule-based algorithmic trading systems. To this end, the study examines the effectiveness of ATR-based take-profit and stop-loss rules as a volatility-adaptive risk-management mechanism within a MACD-based trading framework in the USD/JPY foreign exchange market.

MACD is one of the most widely used trend-following technical indicators (Murphy, 1999; Appel, 1979), while ATR-based exit rules are a commonly adopted approach to volatility-adaptive trade-exit management (e.g., Vezeris et al., 2018). Within this framework, the USD/JPY market, which has recently been characterized by pronounced exchange-rate fluctuations, provides an appropriate setting for evaluating the performance of MACD-based trading systems with ATR-based exit rules under changing market conditions.

This study addresses the limitations identified above in the following ways: (1) by adopting a broad parameter-space design that combines 470 MACD models with unique parameter configurations and 36 ATR-based TP/SL multiplier combinations—enabling a systematic evaluation of a broad set of model and exit-rule configurations; (2) by introducing a classification-based framework that distinguishes multiple distinct outcome patterns before and after the introduction of ATR-based exit rules—enabling the analysis of both performance changes and their underlying mechanisms across different MACD parameter configurations and TP/SL settings; (3) by examining the interaction between MACD parameter configurations, ATR-based exit rules, and market conditions across two distinct subperiods—allowing an assessment of conditional effectiveness under different market conditions.

Together, these design elements constitute a unified analytical framework for evaluating the effectiveness of ATR-based exit rules in algorithmic trading systems. More specifically, the study addresses the following research questions concerning how the effectiveness of volatility-adaptive exit rules depends on model structure, exit-rule design, and market conditions.

1. Overall Effectiveness: *Do volatility-adaptive exit rules improve trading performance across different market conditions?*
The study evaluates trading performance before and after the introduction of ATR-based exit rules across two subperiods.
2. Classification of Diverse Performance Outcomes: *What distinct outcome patterns emerge following the introduction of ATR-based exit rules?*
This question is motivated by the dual role of exit rules in limiting losses and constraining upside gains, which may generate heterogeneous outcomes ranging from

profit enhancement to increased losses. The study classifies outcomes following the introduction of ATR-based exit rules into multiple distinct patterns and examines their distribution.

3. Characteristics of Profit-Enhancement Cases: *What characteristics distinguish cases exhibiting the profit-enhancement pattern, in which already profitable MACD models achieve higher returns following the introduction of ATR-based exit rules?*

The study analyzes the TP/SL multiplier combinations and MACD parameter configurations associated with profit-enhancement cases. This approach clarifies how exit-rule design and the structural characteristics of MACD models jointly shape trading performance outcomes.

4. Market Dynamics and Profit-Enhancement Cases: *How are the observed profit-enhancement cases related to changes in market dynamics?*

The study examines changes in volatility and return autocorrelation across the two sub-periods and evaluates whether profit-enhancement cases are associated with changes in the underlying market dynamics.

5. Fundamental Determinant: *What is the fundamental determinant of performance improvements: the specification of volatility-adaptive exit rules or the underlying trading-model structure?*

The study conducts a controlled comparison under identical ATR-based exit-rule settings and common market conditions by comparing the performance of the standard MACD (12,26,9) model with that of MACD parameter configurations associated with profit-enhancement cases. This approach evaluates whether performance improvements are driven by volatility-adaptive exit rules or arise primarily from appropriately specified MACD configurations, thereby clarifying the role of parameter optimization in shaping trading performance outcomes.

Through this analytical framework, the study seeks to offer broader insights into the role of volatility-adaptive risk-management mechanisms in the design and performance of rule-based algorithmic trading systems. Such a structured approach to decomposing and evaluating the effectiveness of volatility-adaptive exit rules has received little attention in the existing literature. The study does not simply examine whether ATR-based exit rules improve trading performance within MACD-based trading systems. Rather, it investigates how the interaction between volatility-adaptive exit rules, model structure, and market conditions influences trading performance and under what conditions distinct performance outcomes emerge. The findings are expected to contribute to a deeper understanding of adaptive risk-management design under changing market conditions.

2. Methodology

2.1. Moving Average Convergence Divergence (MACD)

The MACD is a technical indicator designed to capture changes in the strength, direction, and momentum of price trends. It is defined by three parameters and expressed in the form $MACD(n_1, n_2, n_3)$, such as $MACD(12, 26, 9)$. The first two parameters determine the MACD line, which is calculated as the difference between the short-term (n_1) and long-term (n_2) exponential moving averages of the price series. The third parameter defines the signal line, which is the exponential moving average (EMA) of the MACD line over n_3 periods. Trading signals are generated based on crossovers between the MACD line and the signal line. A buy (long) signal is generated when the MACD line crosses above the signal line, while a sell (short) signal is generated when the MACD line crosses below the signal line.

The MACD line is defined as:

$$MACD_t = EMA_{n_1}(P_t) - EMA_{n_2}(P_t) \quad (1)$$

where P_t denotes the closing price at time t . The signal line is defined as:

$$Signal_t = EMA_{n_3}(MACD_t) \quad (2)$$

where n_3 denotes the length of the EMA window applied to the MACD series. The EMA of a series is computed recursively as:

$$EMA_{n,t}(X) = \alpha X_t + (1 - \alpha) EMA_{n,t-1}(X) \quad (3)$$

where X_t denotes the value of the series X at time t , and the smoothing parameter is given by $\alpha = 2/(n + 1)$.

A key issue in developing MACD-based trading systems is the selection of parameter values (n_1, n_2, n_3), as these parameters control the timing and frequency of trading signals and, consequently, have a substantial impact on trading performance. When parameter values are not appropriately specified to capture price movements, the MACD, like other technical indicators, can generate false trading signals. Accordingly, appropriate parameter selection becomes an important step in developing MACD-based trading systems.

This issue is particularly important in the present study, which examines the effectiveness of ATR-based exit rules within a MACD-based trading framework. If MACD models with arbitrary parameter configurations were used, it would be difficult to disentangle whether observed performance outcomes are attributable to the ATR-based exit rules or to the underlying MACD structure itself.

To address this issue, the parameter space in this study is constructed based on the methodology developed by Kang (2023, 2026), which was previously applied to identify predefined optimal ranges for MACD parameters. The detailed procedure used to determine these ranges is omitted here for brevity (see Appendix A.1 for a summary of the procedure).¹ The resulting optimal parameter ranges are defined as follows:

- A_{n1} : {15, ..., 20} for the first parameter n_1 ;
- A_{n2} : {20, ..., 27} for the second parameter n_2 ;
- A_{n3} : {16, ..., 25} for the third parameter n_3 .

Using values from these predefined parameter ranges, the following MACD model group is constructed:

- $A_{n1} - A_{n2} - A_{n3}$ (470).

The numbers in parentheses indicate the total number of valid models. The total number of candidate models is $6 \times 8 \times 10 = 480$, based on the sizes of the parameter ranges $n(A_{n1}) = 6$, $n(A_{n2}) = 8$, and $n(A_{n3}) = 10$. Among these, 10 models are excluded because they violate the structural condition $n_1 < n_2$ required for the MACD models, resulting in 470 valid models.

The reason for using a large number of MACD models is to avoid relying on a few narrowly defined sets of parameter configurations. By restricting the analysis to models within predefined optimal parameter ranges, the study aims to isolate the incremental effect of ATR-based exit rules while controlling for their baseline performance characteristics of the trading models. This approach allows for a more precise assessment of the interaction between model structure and exit-rule design.

2.2. Average True Range (ATR)

The Average True Range (ATR) is a standard indicator for measuring market volatility. It is based on the concept of the True Range (TR), which is defined at time t as:

$$TR_t = \max\{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\} \quad (4)$$

where H_t , L_t , and C_{t-1} denote the high and low prices at time t , and the closing price at time $t - 1$, respectively. The ATR is computed using Wilder's smoothing method:

$$ATR_t = \frac{ATR_{t-1} \times (N - 1) + TR_t}{N} \quad (5)$$

where N denotes the ATR period. Higher ATR values indicate more volatile market conditions, whereas lower values correspond to relatively stable periods. This property allows the ATR to be used for dynamically scaling trading thresholds, such as take-profit and stop-loss levels, thereby enabling volatility-adjusted exit rules.

Let E_t denote the entry price at time t , and ATR_t the corresponding ATR value. The stop-loss (SL) and take-profit (TP) thresholds are defined as follows:

$$SL_t = \begin{cases} E_t - k_1 ATR_t & \text{for a long position} \\ E_t + k_1 ATR_t & \text{for a short position} \end{cases} \quad (6)$$

$$TP_t = \begin{cases} E_t + k_2 ATR_t & \text{for a long position} \\ E_t - k_2 ATR_t & \text{for a short position} \end{cases} \quad (7)$$

where k_1 and k_2 are positive multipliers that determine the sensitivity of exit levels to market volatility. Under these rules, an existing position is closed when the market price reaches either the stop-loss or take-profit threshold.

The multipliers k_1 and k_2 play a crucial role in determining the trade-off between risk and return. Larger values of k_1 and k_2 allow for wider price fluctuations and potentially larger gains at the cost of greater exposure to adverse price movements, whereas smaller values result in tighter risk control and more frequent exits.

The ATR period was set to $N = 14$, a commonly used specification for capturing relatively short-term market volatility. The multipliers were defined as $k_1 \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5\}$ and $k_2 \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5\}$. The multiplier range from 1.0 to 3.5 was selected to examine a broad spectrum of volatility-adjusted exit conditions, ranging from relatively tight to wider volatility-adjusted exit thresholds.

The choice of $N = 14$ is further supported by the empirical distribution of ATR (14) over the sample period (2022–2025), as reported in Table A1 of Appendix A.2. The interquartile range remains relatively stable across years, with most observations concentrated between approximately 1.08 and 1.60 yen. These values imply that the selected TP/SL multiplier range corresponds to a sufficiently broad set of volatility-adjusted trading thresholds under different volatility environments.

All combinations of these stop-loss and take-profit multipliers were tested, resulting in a total of 36 SL/TP combinations. Combined with the 470 MACD model specifications described in the previous section, these TP/SL multiplier combinations yield a total of 16,920 cases, each of which is evaluated to assess trading performance.

2.3. Trading Strategy Rules

The trading strategy combines MACD-based entry rules with ATR-based exit rules. Entry signals are generated using the MACD indicator, while exit decisions are primarily determined by ATR-based stop-loss and take-profit thresholds.

A long (short) position is initiated when the MACD line crosses above (below) the signal line. To ensure comparability across model specifications, all trades are executed with a fixed position size of one standardized unit, making the results invariant to scaling. In addition, only one position is held at any point in time. Subsequent signals in the same direction are ignored while an existing position remains open.

All trades are executed at the closing price of the next trading day following signal generation. If an opposite MACD signal is generated before either ATR-based threshold is reached, the existing position is closed, and a new position in the opposite direction is opened at the next trading day's closing price. Accordingly, this trading rule generates alternating long and short positions, where long positions benefit from price increases and short positions benefit from price decreases.

Upon entry, ATR-based SL and TP thresholds are defined using the predetermined multipliers described in Section 2.2. The position is maintained until the market price reaches either threshold under a first-hitting rule, at which point the position is closed at the corresponding level. After a position is closed by either the SL or TP threshold, no new trade is initiated until a new MACD crossover signal occurs. If an opposite MACD crossover signal occurs before either threshold is reached, the position is closed and reversed according to the MACD-based rule, thereby overriding the ATR-based exit mechanism.

This structure separates the roles of the two components of the trading system: MACD rules determine entry and reversal timing through crossover signals, whereas ATR-based rules govern position termination via stop-loss and take-profit thresholds, unless overridden by an opposite MACD crossover signal.

Transaction costs are not incorporated into the analysis, as the objective of the study is to evaluate the effect of ATR-based exit rules on trading performance. Since all model specifications trade a single standardized unit position at a time, transaction costs would apply similarly across alternative model specifications. Excluding transaction costs therefore facilitates a cleaner comparison of relative performance across model specifications.

2.4. Data

This study uses daily exchange rate data for USD/JPY over the period from 1 January 2022 to 31 December 2025, obtained from <http://investing.com/>, a widely used platform for financial market data. The USD/JPY exchange rate is selected as a representative and highly liquid currency pair in the foreign exchange market.

The sample period is selected based on the distinct differences in market behavior observed between the four-year periods before and after 2022, as illustrated in Figure A2 of Appendix A.3. In particular, the post-2022 period exhibits stronger price trends and elevated volatility relative to the preceding period. Such market conditions are particularly relevant for evaluating volatility-based exit rules, as the effectiveness of ATR-based stop-loss and take-profit mechanisms depends on the magnitude and persistence of price fluctuations. Therefore, this study focuses on the 2022–2025 period to examine whether ATR-based exit rules can improve trading performance under relatively volatile market conditions.

The sample period is divided into two segments: the first two years (2022–2023) are used for in-sample testing, while the last two years (2024–2025) constitute the out-of-sample period. This split allows the performance of the 470 MACD models, both before and after the introduction of ATR-based exit rules, to be evaluated within the in-sample period and under out-of-sample conditions, thereby enabling an assessment of the robustness of the observed performance changes.

3. Empirical Results

This section presents the empirical results for the full set of 16,920 cases examined in this study, generated from 470 MACD models and 36 ATR-based take-profit and stop-loss multiplier combinations, as described in Section 2.2. Trading performance is evaluated based on cumulative logarithmic returns derived from trade-level returns calculated between entry and exit points.

3.1. Effects of ATR-Based Exit Rules on Return Distributions

Table 1 reports summary statistics of returns across all MACD models for the two subperiods, enabling a direct comparison of performance before and after the introduction of ATR-based exit rules. Panel [a] presents results prior to ATR-based exit rules, while Panel [b] reports the corresponding results after their introduction.

Table 1. Summary Statistics of MACD Model Performance.

Year	Mean	Median	Std. Dev	Min	Max	Q1 (25%)	Q3 (75%)
Panel [a]: Before the introduction of ATR-based exit rules							
2022–2023	0.1820	0.1893	0.0580	0.0341	0.2570	0.1453	0.2364
2024–2025	0.0720	0.0725	0.0292	−0.0057	0.1409	0.0532	0.0966
Panel [b]: After the introduction of ATR-based exit rules							
2022–2023	0.0524	0.0580	0.0535	−0.1375	0.1655	0.0123	0.0945
2024–2025	0.0127	0.0135	0.0420	−0.1503	0.1490	−0.0139	0.0396

First, Panel [a] indicates that the 470 MACD models constructed from the predefined optimal parameter ranges are overall robust. During the in-sample period (2022–2023), all models yield positive returns, as indicated by the positive minimum value (0.0341), with a mean return of 0.1820. In the out-of-sample period (2024–2025), the mean return remains positive at 0.0720, while the minimum return becomes slightly negative (−0.0057). However, closer inspection of the model-level results indicates that only 14 out of 470 models (3%) generate negative returns. This suggests that the MACD models maintain relatively stable performance even under changing market conditions, providing support for the robustness of the predefined optimal parameter space.

Second, a comparison between Panel [a] and Panel [b] shows that the introduction of ATR-based exit rules is associated with a substantial reduction in average performance. In 2022–2023, the mean return declines from 0.1820 to 0.0524, while in 2024–2025 it decreases from 0.0720 to 0.0127. A similar pattern is observed in the minimum values, which become substantially more negative after the introduction of the exit rules. Overall, these results suggest that ATR-based exit rules can lead to substantially worse outcomes in many cases. Specifically, a case-level examination shows that 6427 out of 16,920 cases (38%) generate negative returns.

Third, despite the overall deterioration in central tendency, the maximum return exhibits a mixed pattern across the two subperiods. While the maximum return decreases in the in-sample period (from 0.2570 to 0.1655), it increases slightly in the out-of-sample period (from 0.1409 to 0.1490). This pattern suggests the existence of performance improvements in a subset of cases, particularly among top-performing cases.

Finally, return levels are substantially higher in 2022–2023 than in 2024–2025 in both panels, suggesting that differences in underlying market dynamics may have influenced overall profitability.

Taken together, these results indicate that while the MACD parameter configurations are broadly robust across the two subperiods, the effectiveness of ATR-based exit rules is not uniform and varies considerably depending on market conditions. These findings call for a more detailed examination of the conditions under which ATR-based exit rules become effective, including the role of TP/SL multiplier combinations, the interaction between exit-rule specifications and underlying MACD parameter configurations, and the influence of market dynamics.

3.2. Classification of Performance Changes

To further investigate the diverse outcomes observed in Table 1, Table 2 classifies all model outcomes based on changes in log returns before and after the introduction of ATR-based exit rules. Each case is assigned to one of six patterns according to the direction and magnitude of performance changes, allowing for a systematic assessment of outcomes and the identification of cases in which ATR-based exit rules improve trading performance.

Table 2. Classification of Performance Changes Before and After the Introduction of ATR-Based Exit Rules.

Pattern	Condition	Interpretation	2022–2023	2024–2025
A	$\text{pre} < 0 \rightarrow \text{post} > 0$	Loss to profit	0	46
B	$\text{pre} > 0 \rightarrow \text{post} > 0, \text{post} > \text{pre}$	Profit enhancement	107	801
C	$\text{pre} < 0 \rightarrow \text{post} < 0, \text{post} < \text{pre} $	Loss reduction	0	2
D	$\text{pre} > 0 \rightarrow \text{post} > 0, \text{post} < \text{pre}$	Profit reduction	13,494	9646
E	$\text{pre} < 0 \rightarrow \text{post} < 0, \text{post} > \text{pre} $	Loss increase	0	456
F	$\text{pre} > 0 \rightarrow \text{post} < 0$	Profit to loss	3319	5969

Note: “Pre” and “Post” denote log returns before and after applying ATR-based exit rules, respectively. The totals in each period sum to 16,920 cases, covering all model–rule configurations.

Several features of the performance-change patterns are noteworthy. First, the majority of cases fall into Pattern D (profit reduction), where returns remain positive but decline after the introduction of ATR-based exit rules. This pattern accounts for 13,494 cases in 2022–2023 and 9646 cases in 2024–2025, indicating that the dominant effect of ATR-based exit rules is a reduction in profitability.

Second, a substantial number of cases are classified as Pattern F (profit to loss), particularly in the out-of-sample period, increasing from 3319 in 2022–2023 to 5969 in 2024–2025. In addition, Pattern E (loss increase) is observed only in the out-of-sample period. These results suggest that ATR-based exit rules tend to exacerbate performance decline under less favorable market conditions, affecting both previously profitable models (Pattern F) and already unprofitable models (Pattern E).

Third, despite these overall negative effects, a non-negligible subset of cases exhibits improved performance. Pattern B (profit enhancement) includes 107 cases in 2022–2023 and 801 cases in 2024–2025, indicating that ATR-based exit rules can enhance returns for certain model–rule combinations, particularly in the out-of-sample period. In contrast, Pattern A (loss recovery) and Pattern C (loss reduction) remain extremely limited, suggesting that ATR-based exit rules rarely transform unprofitable strategies into profitable ones. Importantly, performance improvements are largely confined to already profitable configurations (Pattern B), implying that the effectiveness of ATR-based exit rules depends critically on the underlying model structure.

Overall, the impact of ATR-based exit rules is highly heterogeneous. While the dominant effect is a reduction in returns, a meaningful subset of cases exhibits profit enhancement. The following analysis therefore focuses on Pattern B cases to examine the model–rule combinations under which trading performance improves.

3.3. Magnitude of Performance Improvements

Prior to examining the conditions under which trading performance improves across different model–rule combinations, attention is focused on the profit-enhancement cases (Pattern B) to assess the magnitude and distribution of the associated performance improvements. This serves as a preliminary step to assess whether the observed improvements are sufficiently substantial to warrant further investigation.

Table 3 reports the distribution of return ratios, defined as the ratio of post-ATR to pre-ATR log returns, for profit-enhancement cases (Pattern B). Several features are notable.

Table 3. Distribution of Return Ratios for Profit Enhancement Cases (Pattern B).

Year	<1.0	1.0–1.5	1.5–2.0	2.0–2.5	2.5–3.0	3.0–3.5	≥ 3.5	Total
2022–2023	0	103	4	0	0	0	0	107
2024–2025	0	707	69	17	3	2	3	801

Note: Intervals are left-inclusive and right-exclusive.

First, performance improvements are concentrated in the lower range of return ratios in both subperiods. In 2022–2023, the vast majority of cases (103 out of 107, 96.3%) fall within the 1.0–1.5 interval, indicating relatively modest gains. Only four cases fall within the 1.5–2.0 range, and no cases exceed a ratio of 2.0.

In contrast, the 2024–2025 period exhibits not only a higher frequency of improvements but also a broader distribution of outcomes. While most cases (707 out of 801, 88.3%) remain within the 1.0–1.5 interval, a non-negligible number extend into higher ranges. This contrast may reflect changes in market conditions, a possibility examined more directly in Section 3.6.

Focusing on the upper tail of outcomes in 2024–2025, a relatively small but non-negligible number of cases exhibit strong performance improvements: 94 cases exhibit return ratios of at least 1.5 (i.e., +50% relative to pre-ATR performance), and 25 cases exceed 2.0 (i.e., +100% relative to pre-ATR performance). These results indicate that performance gains are not limited to marginal improvements but can be substantial for a subset of configurations.

Figure 1 illustrates the distribution of return ratios reported in Table 3 in detail, highlighting the thin but meaningful upper tail observed in 2024–2025. These findings motivate a closer examination of TP/SL multiplier combinations in profit-enhancement cases.

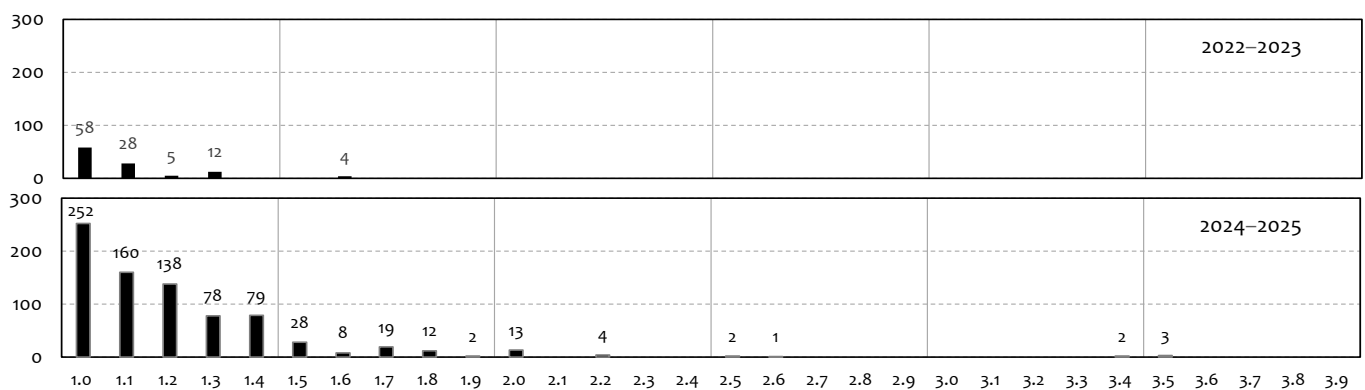


Figure 1. Histogram of Return Ratios for Profit-Enhancement Cases (Pattern B).

3.4. TP/SL Multiplier Combinations in Profit-Enhancement Cases

Which TP/SL multiplier configurations lead to profit enhancement cases (Pattern B)? To address this question, attention is now turned to the distribution of TP/SL multiplier combinations, as shown in Table 4.

Table 4. Distribution of TP/SL Multiplier Combinations in Profit-Enhancement Cases (Pattern B).

		Target-Profit (2022–2023)							Target-Profit (2024–2025)						
		1.0	1.5	2.0	2.5	3.0	3.5	Total	1.0	1.5	2.0	2.5	3.0	3.5	Total
Stop-loss	1.0	–	–	–	–	–	11	11	–	–	1	2	40	15	58
	1.5	–	–	10	1	–	7	18	15	72	9	27	257	25	405
	2.0	–	–	–	–	–	16	16	–	6	–	12	177	8	203
	2.5	–	1	–	–	1	23	25	–	–	–	–	74	1	75
	3.0	–	–	–	–	–	15	15	–	–	–	11	16	–	27
	3.5	–	–	2	–	–	20	22	–	–	–	2	31	–	33
Total		0	1	12	1	1	92	107	15	78	10	54	595	49	801

Note: The table reports the number of cases for each TP/SL multiplier combination. Row and column totals are also reported. A dash (–) indicates that no cases are observed.

First, in the 2022–2023 period, profit-enhancement cases are highly concentrated in combinations involving the highest target-profit multiplier. In particular, a large majority of cases are associated with TP = 3.5 across different stop-loss levels (92 cases, 86.0%). This suggests that profit enhancement during this period was achieved primarily under relatively high profit targets. One possible interpretation is that market conditions allowed for relatively sustained price movements with intermittent increases in volatility, requiring higher TP levels to capture these movements effectively. This interpretation is consistent with Figure A2 in Appendix A.3, which shows sustained but fluctuating price movements in USD/JPY and periods of elevated ATR.

Second, in the 2024–2025 period, a more dispersed pattern is observed. While a substantial concentration remains around higher target-profit multipliers—most notably TP = 3.0 (595 cases, 74.3%)—profit-enhancement cases are distributed across a wider range of TP values, ranging from 1.0 to 3.5. Nevertheless, higher TP levels continue to account for a large share of cases. In addition, certain TP/SL combinations exhibit notable clustering, particularly around TP = 3.0 paired with moderate stop-loss levels such as SL = 1.5 and SL = 2.0. These results suggest that profit enhancement is more likely to occur under configurations combining relatively high profit-target levels with relatively tighter stop-loss levels, highlighting a persistent asymmetry in effective exit rule structures across subperiods.

Third, many TP/SL multiplier combinations yield few or no profit-enhancement cases, particularly those involving lower target-profit multipliers or extreme stop-loss settings. This uneven distribution indicates that the effectiveness of ATR-based exit rules varies substantially across TP/SL multiplier combinations, with profit enhancement concentrated in a limited subset of configurations.

Overall, these findings indicate that profit-enhancement cases are strongly concentrated at higher target-profit levels—TP = 3.5 in 2022–2023 and TP = 2.5–3.5 in 2024–2025—regardless of the stop-loss setting.

This pattern suggests that, under market conditions characterized by relatively sustained price movements and elevated volatility, sufficiently high target-profit multipliers play an important role in generating profit enhancement. However, the observed concentration also indicates that not all TP/SL multiplier combinations are equally effective. Given that the underlying MACD models are constructed from predefined optimal parameter values, profit enhancement appears to occur only under specific TP/SL configurations that are compatible with structurally favorable MACD parameter settings. These findings therefore underscore the importance of examining the MACD parameter configurations associated with profit-enhancement cases.

3.5. MACD Parameter Configurations in Profit-Enhancement Cases

Following the analysis of TP/SL multiplier combinations as external exit rules, the focus shifts to the internal structure of the trading system: which MACD parameter configurations are most frequently associated with profit-enhancement cases (Pattern B)?

The analysis is conducted over the full set of 16,920 cases to address this question. Table 5 reports the number of TP/SL multiplier combinations associated with profit enhancement under each MACD parameter configuration, reflecting how frequently a given configuration generates profit-enhancement outcomes across different exit-rule settings. A non-zero value therefore indicates that at least one ATR-based exit rule is associated with profit enhancement under the corresponding MACD parameter configuration.

Table 5. Distribution of MACD Parameter Configurations in Profit-Enhancement Cases (Pattern B).

2022–2023														2024–2025														
n3														n3														
Group	n1	n2	16	17	18	19	20	21	22	23	24	25	Total	Group	n1	n2	16	17	18	19	20	21	22	23	24	25	Total	
G1	15	20	1	1	2	-	-	-	-	-	-	-	13	G1	15	20	1	2	1	-	-	2	1	1	1	1	2	113
	15	21	1	2	-	-	-	-	-	-	-	-			15	21	2	-	-	-	2	1	1	1	3	3		
	15	22	2	-	-	-	-	-	-	-	-	-			15	22	-	-	-	2	1	1	1	3	4	1		
	15	23	-	-	-	-	-	-	-	-	-	-			15	23	2	-	2	1	1	1	3	4	1	2		
	15	24	-	-	-	-	-	-	-	-	-	-			15	24	1	-	2	1	1	3	4	1	2	2		
	15	25	-	-	-	-	-	-	-	-	-	-			15	25	1	2	1	1	2	3	1	2	2	-		
	15	26	-	-	-	-	-	-	-	-	-	-			15	26	1	1	1	1	3	3	1	2	-	-		
15	27	-	-	-	-	-	-	-	-	-	-	4	15	27	2	1	1	1	3	3	1	3	-	-	-			
G2	16	20	1	2	-	-	-	-	-	-	-	-	20	G2	16	20	-	2	-	-	-	-	-	-	1	3	4	115
	16	21	3	-	-	-	-	-	-	-	-	-			16	21	2	-	-	-	-	-	1	3	4	2		
	16	22	-	-	-	-	-	-	-	-	-	-			16	22	3	-	-	-	-	-	1	3	4	2	2	
	16	23	-	-	-	-	-	-	-	-	-	-			16	23	1	5	1	1	1	3	4	2	2	-		
	16	24	-	-	-	-	-	-	-	-	-	-			16	24	3	6	1	1	3	4	2	2	-	-		
	16	25	-	-	-	-	-	-	-	-	-	2			16	25	2	1	1	3	4	1	2	-	-	-		
	16	26	-	-	-	-	-	-	-	-	-	2			4	16	26	1	1	2	3	1	2	-	-	-		
16	27	-	-	-	-	-	-	-	-	-	4	2	16	27	1	1	3	3	2	2	-	-	-	-				
G3	17	20	2	-	-	-	-	-	-	-	-	-	20	G3	17	20	5	1	-	-	-	-	1	3	3	2	141	
	17	21	-	-	-	-	-	-	-	-	-	-			17	21	3	3	-	-	-	1	3	3	2	2		
	17	22	-	-	-	-	-	-	-	-	-	-			17	22	1	5	-	-	-	1	3	3	2	2		-
	17	23	-	-	-	-	-	-	-	-	-	-			17	23	6	4	4	1	3	3	2	2	-	-		
	17	24	-	-	-	-	-	-	-	-	-	4			17	24	5	5	6	3	4	2	2	-	-	-		
	17	25	-	-	-	-	-	-	-	-	-	2			17	25	5	6	3	4	2	2	-	-	-	-		
	17	26	-	-	-	-	-	-	-	-	-	2			4	17	26	1	2	3	1	2	-	-	-	-		
17	27	-	-	-	-	-	-	-	-	2	2	2	17	27	1	3	1	2	2	2	-	-	-	-				
G4	18	20	-	-	-	-	-	-	-	-	-	-	12	G4	18	20	-	-	-	-	-	1	3	3	2	2	142	
	18	21	-	-	-	-	-	-	-	-	-	-			18	21	2	5	-	-	1	3	3	2	2	-		
	18	22	-	-	-	-	-	-	-	-	-	-			18	22	3	4	2	1	3	3	2	2	-	-		
	18	23	-	-	-	-	-	-	-	-	-	-			18	23	5	4	4	4	3	2	2	-	-	1		
	18	24	-	-	-	-	-	-	-	-	-	-			18	24	5	6	6	5	2	2	-	-	1	-		
	18	25	-	-	-	-	-	-	-	-	-	1			18	25	6	4	4	2	2	-	-	1	-	-		
	18	26	-	-	-	-	-	-	-	-	-	1			3	18	26	2	3	2	2	-	-	1	-	-		4
18	27	-	-	-	-	-	-	-	-	1	3	3	18	27	3	1	2	-	-	-	-	-	-	4	2			
G5	19	20	-	-	-	-	-	-	-	-	-	-	17	G5	19	20	-	-	-	-	1	1	2	2	2	-	142	
	19	21	-	-	-	-	-	-	-	-	-	-			19	21	5	5	-	1	1	2	2	2	-	-		
	19	22	-	-	-	-	-	-	-	-	-	-			19	22	4	2	1	3	2	2	2	-	-	-		
	19	23	-	-	-	-	-	-	-	-	-	-			19	23	5	6	4	5	2	2	-	-	-	-		
	19	24	-	-	-	-	-	-	-	-	-	1			19	24	6	6	5	4	4	-	-	1	1	-		
	19	25	-	-	-	-	-	-	-	-	-	3			19	25	6	6	3	2	-	-	1	-	-	4		
	19	26	-	-	-	-	-	-	-	-	-	3			3	19	26	4	1	2	-	-	1	-	-	4		2
19	27	-	-	-	-	-	-	-	-	3	3	1	19	27	3	2	-	-	-	-	-	4	2	2				
G6	20	20	-	-	-	-	-	-	-	-	-	-	25	G6	20	20	-	-	-	-	-	-	-	-	-	-	148	
	20	21	-	-	-	-	-	-	-	-	-	-			20	21	4	2	1	2	3	5	2	-	-	-		
	20	22	-	-	-	-	-	-	-	-	-	-			20	22	4	1	4	3	5	2	-	-	-	-		
	20	23	-	-	-	-	-	-	-	-	-	3			20	23	6	6	7	4	4	2	-	1	-	-		
	20	24	-	-	-	-	-	-	-	-	-	3			20	24	6	5	4	4	-	3	1	1	-	5		
	20	25	-	-	-	-	-	-	-	-	-	3			3	20	25	5	3	4	1	-	1	-	-	5		1
	20	26	-	-	-	-	-	-	-	-	-	3			1	20	26	4	4	-	-	1	-	-	5	1		2
20	27	-	-	-	-	-	-	-	-	3	1	5	20	27	3	2	-	-	-	-	4	2	2	1				

Note: MACD parameters n_1 and n_2 are arranged along the vertical axis, while parameter n_3 is arranged along the horizontal axis. Each configuration corresponds to a MACD model defined by a specific combination of MACD parameter values (n_1, n_2, n_3). Each subperiod contains 480 parameter configurations, including all 470 MACD models examined in this study. The remaining 10 configurations ($n_1 = n_2 = 20$) are shown for completeness but are excluded from the analysis because they violate the structural condition $n_1 < n_2$, as described in Section 2.1.

A key feature of the table is the distribution of non-zero values, which reveals a marked difference between the two subperiods. In 2022–2023 (left panel), profit-enhancement cases are highly concentrated and unevenly distributed across a limited subset of MACD parameter configurations, suggesting that improvements are confined to very specific regions of the parameter space. By contrast, in 2024–2025 (right panel), non-zero values are

more widely distributed across groups G1–G6 and across different combinations of n_1 , n_2 , and n_3 , indicating that profit enhancement is less concentrated within narrow regions of the parameter space.

To quantify this difference, the number of MACD parameter configurations with non-zero values is counted. In 2022–2023, only 45 parameter configurations meet this condition, corresponding to 9.6% of the 470 MACD models. In contrast, in 2024–2025, this number increases sharply to 313 parameter configurations, corresponding to 66.6% of all 470 MACD models. This substantial expansion indicates a shift from highly concentrated to more widely distributed profit-enhancement patterns across MACD parameter settings in the later period.

Figure 2 provides a visual representation of profit-enhancement cases (Pattern B) across MACD parameter configurations. The vertical axis represents combinations of n_1 and n_2 , which determine the MACD line, while the horizontal axis represents n_3 , corresponding to the signal-line parameter. Color intensity reflects the number of TP/SL multiplier combinations associated with profit enhancement under each MACD parameter configuration. This visualization facilitates a more direct assessment of the structural patterns identified in Table 5 and more clearly reveals the spatial concentration, clustering, and intensity of profit-enhancement cases across the MACD parameter space.

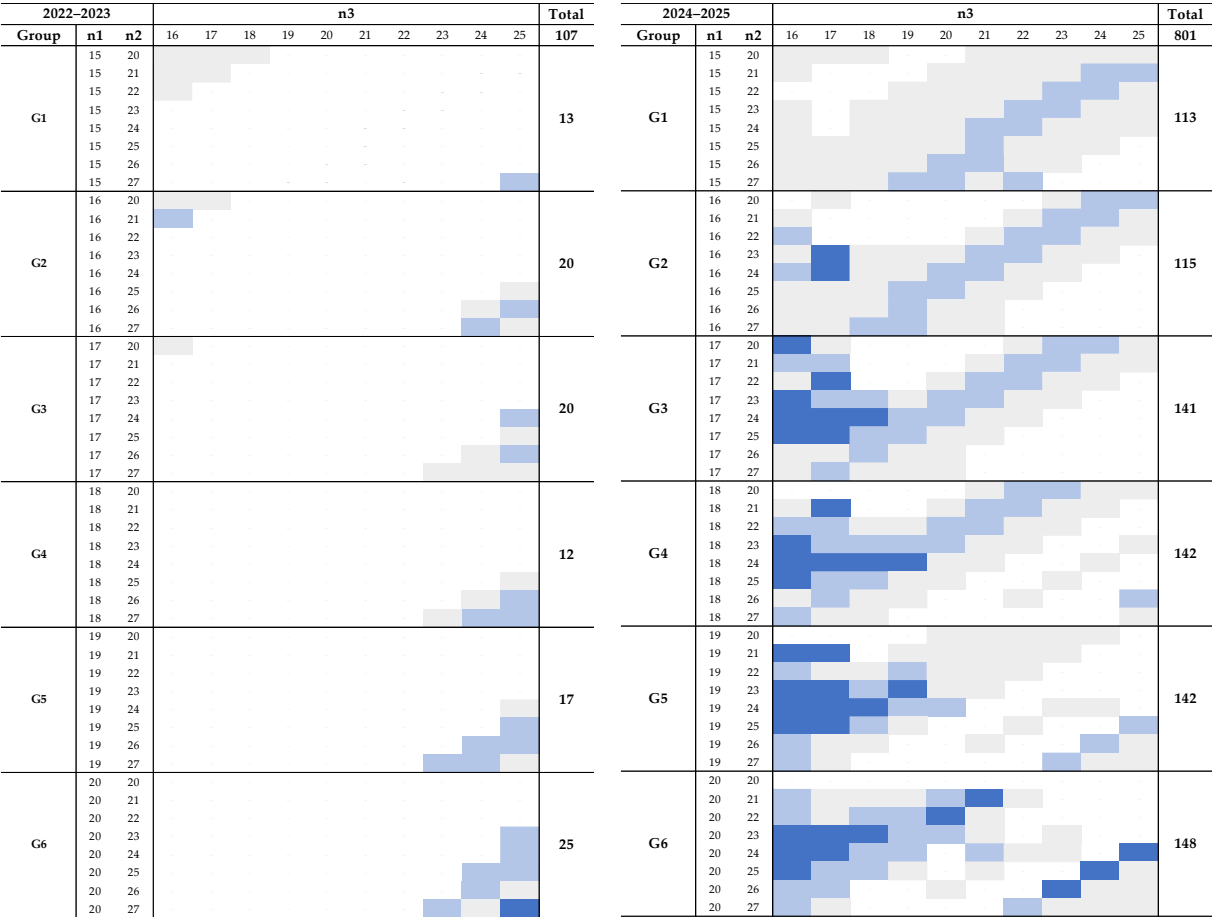


Figure 2. Heatmap of Profit-Enhancement Cases (Pattern B) across MACD Parameter Configurations. Note: Color intensity reflects the number of cases reported in Table 5. White indicates zero cases, gray indicates 1–2 cases, light blue indicates 3–4 cases, and dark blue indicates 5 or more cases.

The contrast between the two panels is immediately apparent. In 2022–2023 (left panel), profit enhancement is confined to a relatively small number of isolated configurations, with low intensity and little evidence of clustering. By contrast, in 2024–2025 (right panel), both

the coverage and intensity of profit enhancement increase markedly, with multiple regions exhibiting visibly higher concentrations.

Notably, the right panel shows that high-intensity regions (dark blue) are not randomly distributed but instead reflect an underlying structure within the MACD parameter space. Darker cells indicate MACD parameter configurations associated with a larger number of TP/SL multiplier combinations that are linked to profit enhancement. This pattern suggests that, for those configurations, the effectiveness of ATR-based exit rules is not limited to a single finely tuned TP/SL setting but extends across multiple TP/SL multiplier combinations.

Moreover, it is worth noting that the high-intensity cells in the right panel are concentrated at relatively small values of the signal-line parameter n_3 , rather than being evenly distributed across the horizontal axis. This systematic pattern provides additional evidence that profit enhancement is associated with specific regions of the MACD parameter space. In this sense, the figure suggests that the underlying MACD parameter configuration may play a more important role in determining profit enhancement than the precise choice of TP/SL multipliers.

Taken together, these patterns indicate that profit enhancement is systematically associated with specific regions of the MACD parameter space rather than arising randomly across configurations, highlighting the importance of the underlying MACD parameter structure in shaping the effectiveness of ATR-based exit rules.

3.6. Market Dynamics and Performance Improvements

To what extent can the observed expansion in profit-enhancement cases be explained by changes in market dynamics? Table 6 reports summary measures of volatility and return autocorrelation, providing a basis for examining whether these observed performance patterns are associated with changes in the underlying market dynamics.

Table 6. Market Dynamics of USD/JPY: Volatility and Return Autocorrelation.

Panel (a): Annualized 14-Day Rolling Volatility							
Year	Mean	Median	Std. Dev	Min	Max	Q1 (25%)	Q3 (75%)
2022	0.1075	0.1050	0.0436	0.0398	0.2310	0.0753	0.1309
2023	0.0968	0.0925	0.0312	0.0366	0.2063	0.0746	0.1175
2024	0.0940	0.0899	0.0338	0.0221	0.1735	0.0715	0.1168
2025	0.0947	0.0857	0.0297	0.0461	0.1687	0.0719	0.1134
Panel (b): 14-Day Rolling Lag-1 Autocorrelation							
Year	Mean	Median	Std. Dev	Min	Max	Q1 (25%)	Q3 (75%)
2022	−0.0727	−0.1021	0.1894	−0.7008	0.4820	−0.1854	0.0329
2023	−0.0547	−0.0697	0.2616	−0.5372	0.5109	−0.2710	0.1390
2024	−0.0786	−0.1404	0.2719	−0.6586	0.5234	−0.2759	0.1283
2025	−0.1624	−0.1636	0.2200	−0.7896	0.4820	−0.2746	−0.0593

Note: Panel (a) reports annualized volatility computed from the standard deviation of daily log returns over a 14-day rolling window, based on 250 trading days per year. Panel (b) reports the corresponding lag-1 autocorrelation of daily log returns computed over the same 14-day rolling window.

The results indicate that volatility does not exhibit a strictly monotonic trend, but shows a generally declining pattern from 2022 to 2025. This pattern suggests gradual variations in market conditions rather than a pronounced structural shift. More importantly, the autocorrelation measures remain predominantly negative across all years, with more pronounced negative values observed in 2025. This pattern suggests the presence of mild mean-reverting dynamics rather than persistent trend-following behavior.

This evidence indicates that the increase in profit-enhancement cases (Pattern B) in the later period cannot be explained by stronger trend persistence. Rather, it is consistent with a market environment characterized by more pronounced short-term reversals, in which prices tend to revert over short horizons. Under such conditions, volatility-based exit rules may enhance performance by adapting exit timing to short-term price movements, allowing profitable positions to persist while limiting adverse price movements at an early stage.

Further support for this interpretation is provided by the distribution of TP/SL multiplier combinations in the 2024–2025 period, as reported in Table 4. Profit-enhancement cases are concentrated around TP = 3.0 combined with moderate stop-loss levels, particularly SL = 1.5 and SL = 2.0. This pattern suggests that allowing sufficient room for profit-taking while applying relatively tighter loss limits contributes to improved performance.

Taken together, these findings provide a coherent explanation for the expansion observed in Figure 2. Under mildly mean-reverting market conditions, a broader set of structurally favorable MACD parameter configurations derived from predefined optimal parameter values appears to become compatible with ATR-based exit rules, resulting in both greater coverage and higher intensity of profit-enhancement cases. Consequently, these findings suggest that the interaction between model structure and ATR-based exit rules is a key factor in explaining the observed performance improvements.

3.7. The Importance of Parameter Optimization

The preceding analysis highlights that profit enhancement arises through the interaction between market conditions, ATR-based exit rules, and the underlying MACD parameter structure. This raises a more fundamental question: between ATR-based exit rules and MACD parameter structure, which plays the more dominant role in shaping performance outcomes?

To address this issue, this subsection conducts a controlled comparison under common market conditions and the same ATR-based exit-rule frameworks, allowing attention to focus on differences in MACD parameter configurations. Specifically, the performance of the standard MACD (12,26,9) model is compared with that of the MACD models associated with Pattern B cases reported in Table 5. Because both groups are evaluated under the same market conditions and ATR-based exit-rule framework, any substantial difference in performance can be interpreted primarily as reflecting differences in MACD parameter configurations, thereby providing evidence on the importance of parameter optimization.

Table 7 reveals a clear contrast between the standard MACD model and the MACD models associated with Pattern B cases. The standard model yields negative average returns in both subperiods, whereas the MACD models associated with Pattern B cases exhibit consistently positive returns. This difference extends beyond the mean, as the entire distribution for the latter group is shifted upward, with higher medians and positive lower quartiles. In addition, return dispersion is smaller for these models, indicating more stable performance.

Table 7. Return Distribution of Standard MACD Model and MACD Models with Pattern B Cases.

	Period	N	Mean	Median	Std. Dev	Min	Max	Q1 (25%)	Q3 (75%)
Standard MACD (12,26,9)	2022–2023	36	−0.0476	−0.0398	0.0564	−0.1801	0.0535	−0.0775	−0.0048
	2024–2025	36	−0.0327	−0.0341	0.0588	−0.1367	0.0965	−0.0826	0.0004
MACD Models (Pattern B Cases)	2022–2023	107	0.0839	0.0895	0.0242	0.0357	0.1193	0.0614	0.1038
	2024–2025	801	0.0869	0.0870	0.0227	0.0526	0.1305	0.0664	0.1069

Note: N denotes the number of cases. The standard MACD (12,26,9) represents a single benchmark model, including all 36 cases generated under combinations of TP/SL multiplier settings. The 107 Pattern B cases in 2022–2023 were generated from 45 unique MACD models, whereas the 801 Pattern B cases in 2024–2025 were generated from 313 unique MACD models under the same TP/SL multiplier framework.

Taken together, these results provide strong evidence that parameter optimization plays a central role in determining the effectiveness of ATR-based exit rules, as applying such rules to the standard MACD (12,26,9) specification fails to deliver comparable improvements.

This interpretation is reinforced by the classification of outcomes for the standard MACD model under 36 combinations of TP/SL multiplier settings. In both subperiods, the majority of cases fall into Pattern E (loss increase), with no instances of Pattern B (profit enhancement); see Table A2 in Appendix B. This suggests that ATR-based exit rules may not generate performance improvements when applied to models with non-optimized parameter configurations, and that their effectiveness depends critically on the selection of MACD parameter configurations. Notably, the standard MACD (12,26,9) specification lies outside the optimal parameter values identified in Section 3.2.

To further clarify why optimized MACD parameter configurations produce different performance outcomes, the following analysis examines differences in trading behavior. While Table 7 demonstrates that parameter optimization plays a central role in the effectiveness of ATR-based exit rules, it does not fully explain how these differences emerge in actual trading activity and trade outcomes. Table 8 therefore compares trading activity and win rates between the standard MACD model and the MACD models associated with Pattern B cases.

Table 8. Trading Activity and Win Rates: Standard MACD model vs. MACD Models Associated with Pattern B Cases.

	Period	N	Avg. Trades	Win Rate	Long	Short
Standard MACD (12,26,9)	2022–2023	36	41.56	0.388	0.490	0.291
	2024–2025	36	36.81	0.380	0.495	0.268
MACD Models (Pattern B Cases)	2022–2023	107	22.15	0.463	0.657	0.289
	2024–2025	801	21.24	0.496	0.499	0.496

Note: N denotes the number of cases. Avg. Trades denotes the average number of trades per case. Win Rate, Long, and Short denote the average proportions of winning trades for all trades, long positions, and short positions, respectively.

The results reveal substantial contrasts in both the frequency and effectiveness of trades. The standard MACD (12,26,9) model exhibits a substantially larger number of trades, yet relatively low win rates, particularly for short positions. In contrast, the MACD models associated with Pattern B cases generate substantially fewer trades (22.15 vs. 41.56 and 21.24 vs. 36.81, respectively) while achieving higher win rates, indicating more selective and effective trading behavior. Since fewer trades generally imply lower transaction costs, incorporating real transaction costs (e.g., bid–ask spreads) would likely further enhance their practical advantage over the standard MACD model.

Moreover, the standard MACD model displays a pronounced asymmetry between long and short performance, with short trades consistently underperforming. This imbalance is largely mitigated in the MACD models associated with Pattern B cases, especially in the 2024–2025 period, where win rates become more balanced across positions.

Taken together, these findings indicate that the superior performance of the MACD models associated with Pattern B cases is reflected not only in higher returns, but also in structurally improved trading behavior—characterized by reduced overtrading, more effective exit management, and more balanced performance across market directions. Consistent with the upward shift in return distributions observed in Table 7, these results further reinforce the view that parameter optimization plays a central role in determining the effectiveness of ATR-based exit rules.

Finally, a close examination of the MACD models associated with Pattern B cases provides further insight into the role of the ATR-based exit mechanism. Without ATR-based

TP/SL rules, these models generate only 12.23 trades on average in 2022–2023 and 10.70 in 2024–2025. These values are substantially lower than those reported in Table 8 for the corresponding ATR-based models (22.15 and 21.24, respectively). This finding suggests that the superior performance of the Pattern B cases is not primarily attributable to positions remaining open until MACD reversal signals occur. Rather, the introduction of ATR-based TP/SL rules increases the frequency of TP/SL-triggered position-closing events, indicating that the adaptive exit mechanism actively contributes to more effective trade management by realizing profits or limiting losses before subsequent MACD reversal signals occur.

4. Concluding Remarks

This study examined ATR-based take-profit and stop-loss rules as components of adaptive risk-management design in algorithmic trading systems. The objective was not simply to evaluate the effectiveness of volatility-adaptive exit rules, but to identify the conditions under which interactions between exit-rule design, trading-model structure, and market conditions contribute to robust system performance. The findings demonstrate that the effectiveness of such rules depends on the interaction between exit-rule design, trading-model structure, and market conditions.

4.1. Answers to the Research Questions

The empirical findings provide systematic answers to the five research questions presented in the Introduction.

For the first research question, the results show that ATR-based exit rules do not generate uniform improvements across all market conditions or model structure. The underlying MACD models constructed using predefined optimal parameter values exhibit broadly robust performance across the two subperiods. Even within this favorable modeling environment, however, performance improvements are observed only in a subset of the evaluated cases, while many model configurations experience deteriorating profitability or increased losses following the introduction of ATR-based exit rules. These findings suggest that the effectiveness of volatility-adaptive exit rules depends critically on their compatibility with both model structure and market conditions.

For the second research question, the classification analysis clarifies why this conditional effectiveness arises. Performance outcomes are highly heterogeneous, with profit reduction (Pattern D) representing the dominant outcome in both subperiods. However, performance improvements arise primarily from the enhancement of already profitable model configurations (Pattern B) rather than from transforming unprofitable configurations into profitable ones.

For the third research question, profit-enhancement cases are systematically associated with specific TP/SL multiplier combinations and optimized MACD parameter configurations. In particular, improved outcomes are concentrated at relatively high take-profit levels combined with moderate stop-loss thresholds, indicating an asymmetric exit structure that allows profits to run while maintaining relatively tight loss limits. At the same time, these effects are not driven by TP/SL settings alone, but depend critically on the compatibility between TP/SL settings and underlying MACD parameter configurations. The results further show that profit-enhancement cases are concentrated in specific regions of the MACD parameter space, with some configurations generating improvements across multiple TP/SL combinations, suggesting that within these regions, effectiveness is robust across multiple TP/SL multiplier combinations rather than confined to a single one.

For the fourth research question, the expansion of profit-enhancement cases in the 2024–2025 period appears to be closely related to changes in market dynamics. The evidence suggests that this expansion cannot be attributed to stronger trend persistence.

Rather, it appears that mildly mean-reverting market conditions, characterized by more pronounced short-term reversals, enhance the compatibility between ATR-based exit rules and optimized MACD parameter configurations, thereby expanding both the coverage and intensity of profit-enhancement cases.

For the fifth research question, the results indicate that parameter optimization plays a central role in determining the effectiveness of ATR-based exit rules. A comparison with the standard MACD (12,26,9) model provides clear evidence of the importance of model specification. The standard configuration, which lies outside the predefined optimal parameter regions, generates no profit-enhancement cases and is predominantly associated with loss-increase outcomes. This contrast demonstrates that ATR-based exit rules may not generate performance improvements when applied to models with non-optimized parameter configurations, but tend to become effective primarily when combined with appropriately specified MACD parameter configurations. Consistent with this interpretation, the MACD models associated with profit-enhancement cases exhibit fewer trades, higher win rates, and more balanced performance across long and short positions, indicating more selective trade execution and more effective trade management. These findings also suggest that parameter optimization may contribute to improved risk characteristics, such as reduced maximum drawdown and enhanced risk-adjusted performance, although these measures were not examined in the present study.

In conclusion, the findings demonstrate that the effectiveness of ATR-based exit rules depends critically on their compatibility with underlying model structures and market conditions. Rather than generating universally positive effects, ATR-based exit rules improve performance only under specific conditions characterized by compatibility between appropriately specified trading models, suitable TP/SL multiplier combinations, and prevailing market dynamics. In this context, volatility-adaptive exit rules function primarily as a complementary mechanism that reinforces trading structures rather than independently generating superior outcomes.

4.2. Methodological Features of the Study

Several aspects of the research design are important for interpreting the findings of the present study.

(1) Predefined Optimal Parameter Ranges

A fundamental methodological feature of the study is the construction of MACD trading models from predefined optimal parameter ranges. Following the methodology proposed by Kang (2023, 2026), the analysis is restricted to MACD parameter configurations that satisfy predefined optimality criteria. This design allows for the isolation of the effects of ATR-based exit rules from those of the underlying MACD model structure.

(2) Broad Trading-System Design

A second methodological feature is the adoption of a broad trading-system design combining 470 MACD parameter configurations with 36 ATR-based TP/SL multiplier combinations, yielding a total of 16,920 cases. This design enables a systematic examination of how model structure and exit-rule design jointly determine trading performance.

(3) Classification-Based Approach

A third methodological feature is the classification-based approach used to evaluate performance changes before and after the introduction of ATR-based exit rules. By classifying performance changes into multiple outcome patterns, the framework provides a more nuanced understanding of how ATR-based exit rules affect trading outcomes.

(4) Controlled Comparison

A final methodological feature is the controlled comparison between the standard MACD (12,26,9) model and MACD models associated with profit-enhancement cases. By holding the trading framework and ATR-based exit framework constant, this comparison isolates the role of parameter optimization, thereby clarifying the importance of model specification in determining the effectiveness of ATR-based exit rules.

Taken together, these methodological features make it possible to identify the conditional effectiveness of volatility-adaptive exit rules and to evaluate the relative importance of parameter optimization and exit-rule design within algorithmic trading systems.

4.3. Contributions to the Literature

This study contributes to the literature by providing new insights into the conditional effectiveness of volatility-adaptive exit rules.

[Kaminski and Lo \(2014\)](#) show that the effectiveness of stop-loss rules depends critically on the underlying return process. In particular, stop-loss rules always reduce expected returns under a random walk. Under mean-reverting conditions, stop-loss policies can miss subsequent reversals after cumulative losses and thereby lower expected returns.

The market-dynamics analysis conducted in the present study reveals evidence of mild mean-reverting behavior in the later subperiod. Interestingly, this period also exhibits a substantial expansion of profit-enhancement cases.

This apparent contrast arises because [Kaminski and Lo \(2014\)](#) examine stand-alone stop-loss rules, whereas the present study evaluates a volatility-adaptive exit mechanism that jointly specifies take-profit and stop-loss levels. During this period, profit-enhancement outcomes are observed across a broader range of MACD parameter configurations under the ATR-based exit mechanism. In this sense, the present study extends the perspective of [Kaminski and Lo \(2014\)](#) by showing that the effectiveness of exit rules is determined not solely by the stochastic properties of returns, but also by the interaction between model structure, exit-rule design, and market conditions.

The findings of the present study also provide an important point of comparison with [Clare et al. \(2013\)](#), who examine moving-average breakout stop-loss rules and trailing stop-loss rules based on fixed-percentage declines from the entry price. Across both mechanisms, they find little evidence that stop-loss rules improve performance and conclude that “a change of trend is the best stop loss.”

The findings of the present study are broadly consistent with this conclusion insofar as ATR-based exit rules do not generate uniformly positive effects. However, important differences in research design lead to a more nuanced interpretation. Whereas [Clare et al. \(2013\)](#) focus on long-only trend-following strategies, the present study evaluates both long and short positions within a broad parameter-space framework. As a result, the broader analytical framework adopted in the present study makes it possible to identify profit-enhancement cases that might otherwise remain undetected.

4.4. Research and Practical Implications

For researchers, the findings of the present study suggest that broad parameter-space analysis and classification-based approaches can reveal the conditions under which adaptive risk-management mechanisms become effective. Future research may extend this framework to other asset classes, trading strategies, and risk-management mechanisms.

For practitioners, the findings of the present study provide guidance regarding ATR-based TP/SL multiplier combinations and MACD parameter configurations associated with profit-enhancement outcomes in the USD/JPY market. More broadly, the results suggest that exit-rule design should be viewed as a complementary component of system optimization rather than an independent source of superior performance. More funda-

mentally, the results indicate that appropriate parameter optimization of the underlying trading model represents the primary determinant of performance, while exit-rule design plays a secondary role. The comparison with the standard MACD (12,26,9) model shows that ATR-based exit rules fail to generate any profit-enhancement cases when applied to a trading model with non-optimized parameter configurations. This finding highlights the importance of parameter optimization when evaluating alternative risk-management mechanisms.

4.5. Limitations of the Study

The findings should be interpreted in light of several limitations. First, the analysis focuses on a single currency pair (USD/JPY) and a specific ATR-based exit-rule design. Second, the study examines ATR-based exit rules within a MACD-based trading framework, and the findings may not generalize directly to alternative trading-model structures or risk-management mechanisms. Future research may extend the present framework to different asset classes, market environments, and trading systems to further evaluate the robustness and generalizability of the results.

One limitation of the present framework is that ATR is calculated using a 14-day smoothing window. Consequently, sudden market shocks, such as central bank currency interventions that drastically increase intraday and daily volatility, may not be fully reflected in the ATR value immediately. This represents an inherent limitation of ATR-based volatility measures when responding to abrupt market shocks. It should be noted, however, that unlike classic fixed stop-loss rules, subsequent trades can incorporate an updated ATR value, allowing the framework to gradually adapt to changes in market volatility.

It is also worth noting that the TP/SL multiplier range (1.0–3.5) adopted in this study was determined empirically for the USD/JPY market over the 2022–2025 period and does not imply that the same TP/SL multiplier range is necessarily appropriate for other financial markets or different sample periods. The optimal TP multiplier is likely to depend on market conditions. Accordingly, the appropriate TP/SL multiplier range should be determined based on the characteristics of the financial market and the sample period under investigation.

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Appendix A. Supplementary Methodological Materials

Appendix A.1. Procedure for Determining the MACD Parameter Ranges

The MACD parameter ranges are determined based on the methodology proposed by Kang (2023, 2026), as follows:

- *Step 1: Setting the Parameter Search Space and Evaluating Sample Models*

In this study, the parameter search space is defined as $n_1 = \{3, \dots, 20\}$, $n_2 = \{5, \dots, 40\}$, and $n_3 = \{3, \dots, 40\}$, with each parameter incremented by one day. These ranges are selected to include MACD models representing both relatively short-term trading strategies characterized by frequent trading signals, and relatively longer-term trading strategies with less frequent signals. This search space covers MACD specifications ranging from MACD (3, 5, 3) to MACD (20, 40, 40) and yields 24,624 ($=18 \times 36 \times 38$) models with distinct parameter combinations. After excluding models that violate the MACD structural condition $n_1 < n_2$, the remaining 19,456 models constitute the baseline sample used for the parameter-range selection procedure.

Using USD/JPY foreign exchange market data for the 2022–2023 period, trading simulations are conducted for all models within the baseline sample to evaluate trading performance. Based on the simulation results, the top 100 performing models are identified and retained for subsequent analysis.

- *Step 2: Examining Parameter Distributions*

For the top 100 ranked models, the frequency distributions of the three parameter values (n_1 , n_2 , n_3) are examined. Parameter values whose frequency ratios exceed a predetermined cutoff point are identified as predefined optimal values associated with higher trading performance. In this study, the cutoff point is set at 5%. Figure A1 shows the frequency distributions of the three parameter values among the top 100 models. For ease of exposition, Figure A1 is presented at the end of this section.

- *Step 3: Identifying Optimal Parameter Ranges and Constructing New Sample Models*

For the parameter values identified in Step 2, attention is focused on consecutive values clustered around the most frequently observed parameter values. These values are considered to form predefined continuous optimal parameter ranges. New sample models are then constructed by combining parameter values within the resulting ranges for the three MACD parameters. The resulting predefined optimal parameter ranges yield a total of 470 MACD models, as described in Section 2.1.

For reference, 67 of the original top 100 models are included in the resulting set of 470 models, accounting for 13.3% ($=67/470$) of the constructed models. This indicates that most of the 470 models are newly generated parameter combinations derived from the predefined parameter ranges.

- *Step 4: Validating the Performance of the New Sample Models*

Although the newly constructed MACD models are generated from predefined optimal parameter ranges, it remains necessary to verify whether they also produce superior trading performance. If models constructed from these ranges consistently produce superior trading performance across both the in-sample and out-of-sample periods, the predefined parameter ranges can be regarded as valid parameter ranges for the tested market. The empirical results are presented in Section 3.

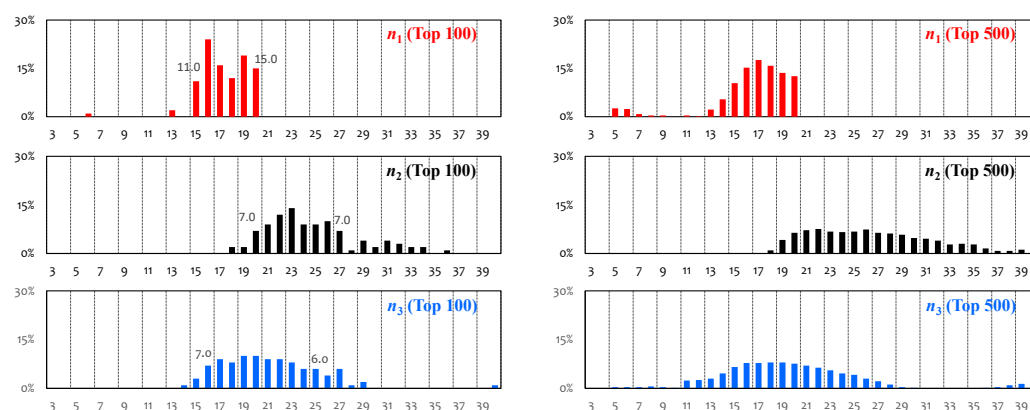


Figure A1. Frequencies of the three parameter values (n_1 , n_2 , n_3) in the top 100 and top 500 models. Note: Figure A1 illustrates the frequency distributions of the three parameter values among the top 100 and top 500 performing models selected from the 19,456 baseline models. Extending the analysis from the top 100 to the top 500 models reveals no substantial changes in the overall distributional structure. This suggests that the top 100 models already capture the characteristic patterns of higher-performing MACD models, while remaining broadly consistent with those observed in the top 500 models. This supports the use of the top 100 models for identifying predefined optimal parameter ranges. The numerical values shown on the histogram bars for the top 100 models represent the corresponding frequency ratios.

Appendix A.2. ATR Period Selection and Characteristics of ATR (14)

The ATR period was set to $N = 14$ in order to balance sensitivity to short-term price fluctuations against the ability to capture persistent volatility conditions observed during the sample period. Shorter ATR periods may become overly sensitive to transient market movements, whereas longer periods may smooth out the elevated volatility observed in the post-2022 USD/JPY market. Table A1 indicates that the interquartile range of ATR (14) remains relatively stable across years, suggesting that the selected ATR period captures persistent volatility conditions while avoiding excessive sensitivity to short-term fluctuations.

Table A1. Descriptive Statistics of ATR (14) for USD/JPY (2022–2025).

Year	Mean	Median	Std. Dev	Min	Max	Q1 (25%)	Q3 (75%)
2022	1.37	1.35	0.42	0.64	2.29	1.17	1.64
2023	1.34	1.29	0.29	0.74	2.00	1.16	1.56
2024	1.40	1.40	0.37	0.76	2.29	1.08	1.60
2025	1.39	1.35	0.24	1.00	2.13	1.18	1.52

Note: ATR (14) denotes the 14-period Average True Range calculated from USD/JPY price data. As a price-based volatility measure used to scale TP/SL thresholds in the ATR-based framework, the distribution of ATR (14) provides an empirical basis for selecting the TP/SL multiplier range adopted in this study (1.0–3.5). For example, ATR values around 1.3–1.4 yen imply that $1.0 \times \text{ATR}$ corresponds to approximately 1.3–1.4 yen, representing relatively tight trading thresholds, whereas $3.5 \times \text{ATR}$ corresponds to approximately 4.7–4.9 yen, representing substantially wider volatility-adjusted thresholds.

Appendix A.3. Market Conditions and Sample Period Selection

Figure A2 illustrates the market conditions underlying the selection of the 2022–2025 sample period for testing. Differences in both price dynamics and volatility patterns are observed between the four-year periods before and after 2022. In particular, the post-2022 period is characterized by a pronounced upward trend in the USD/JPY exchange rate, together with elevated ATR levels. This suggests a shift in market dynamics after 2022, marked by stronger trends and increased volatility.

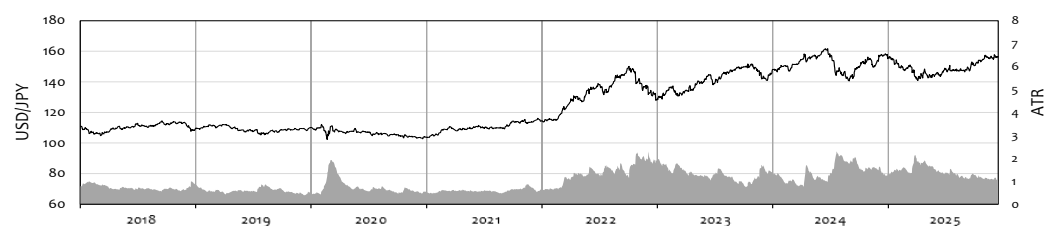


Figure A2. USD/JPY Exchange Rate and ATR (14), 2018–2025. Note: This figure illustrates the evolution of the USD/JPY exchange rate together with ATR (14) over the period from 2018 to 2025. The upper panel shows the exchange rate, while the lower panel shows ATR (14), which captures market volatility. Elevated volatility in the post-2022 period is sustained over time, in contrast to the temporary volatility spike associated with the COVID-19 shock in early 2020.

Appendix B. Classification of Performance Changes for the Standard MACD Model

Table A2 reports the classification of performance changes for the standard MACD (12,26,9) model before and after the introduction of ATR-based exit rules.

Table A2. Classification of Performance Changes for the Standard MACD (12,26,9) Model.

Pattern	Condition	Interpretation	2022–2023	2024–2025
A	pre < 0 → post > 0	Loss to profit	6	9
B	pre > 0 → post > 0, post > pre	Profit enhancement	0	0
C	pre < 0 → post < 0, post < pre	Loss reduction	5	0
D	pre > 0 → post > 0, post < pre	Profit reduction	0	0
E	pre < 0 → post < 0, post > pre	Loss increase	25	27
F	pre > 0 → post < 0	Profit to loss	0	0

Note

- ¹ Kang (2023) applied this methodology to identify optimal MACD parameter ranges for the Nikkei, Dow Jones, and Nasdaq stock-index futures markets, whereas Kang (2026) applied the same methodology to the gold market.

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